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Project Title:	Three-Dimensional Structures and Interface Dynamics in Univariant and Invariant Eutectic Solidification in a Ternary Alloy		
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Comments:			
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Flight Program:	ISS		
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Task Description:

The diverse and expansive spectrum of growth morphologies that are attainable through solidification processing of ternary and higher order systems provides a promising avenue for accessing new structures, materials, and devices with enhanced functionality. However, the challenge of controlling the complex interplay between coupled-phase topology, local defect structures, and non-planar growth modes requires a more complete understanding of the mechanisms of pattern formation and the influence of applied external forces. Focusing on multiphase microstructures in Al-Cu-Ag ternary alloys, this experimental program is aimed at understanding and quantifying the 3D relationships between the relevant structural features, including local eutectic phase topology, crystallographic orientation/misorientation, eutectic fault configuration, faulted domain structure, and grain structure, all relative to the constraints of specimen geometry and applied thermal field.

SIGNIFICANCE

Multiphase multi-scale material structures offer great promise for discoveries of as yet unseen fundamental material responses, new types of device functionality, and improved levels of materials performance. Indeed, innovations in functional heterostructures have led to great developments in photovoltaics, photonics, and even metamaterials. Moreover, limitations to future applications of these and other multiphase materials arise from difficulties in exerting a high level of control over various structural features. As in situ or natural processing avenues to such materials continue to be explored, it is clear that multiphase solidification is one very important pathway toward these and other remarkable materials. The objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled multiphase structures through advanced solidification processing. Moreover, flight-based experiments are critical to both of these areas, serving (i) to provide an environment for fundamental measurement of dynamics that cannot be effectively observed in ground-based studies and (ii) as a potential environment for the fabrication of novel materials.

CRITICAL QUESTIONS:

• What are the geometric and topological principles governing the distribution of phases, fault domains, lamellar fault structure, grain structure, and interface morphology? Further, how do these depend on specific crystal orientations with respect to (i) neighboring lamellae, (ii) overall periodicity, (iii) the directional growth axis, and (iv) the thermal gradient and specimen geometry?

• How are the geometric relationships correlated with pattern formation dynamics and the selection of specific growth morphologies? Specifically, how do the basic structural and crystallographic features give rise to (i) lamellar fault formation and ordering, (ii) the maze-lamellae transition, (ii) eutectic cells and dendrites, and (iv) selection of 3-phase structures?

• How do the dynamics respond to thermal and geometric constraints that could be applied for the purpose of controlling the structure?

OVERARCHING PRINCIPAL OBJECTIVES:

• Establish an analytical framework for 3D quantitative description of the multiphase microstructures, combining geometric and crystallographic parameters for phase boundaries, domain boundaries (faults), grain boundaries, and the solid-liquid interface itself.

• Understand the local structural interactions and fundamental mechanisms that govern morphological dynamics and the evolution of complex 3D multiphase microstructures.

• Employ the newly developed quantitative framework and mechanistic understanding to investigate the collective dynamics of pattern formation, specifically examining steady-state pattern dynamics, the natural transient under directional growth conditions, the development of non-planar growth modes along the univariant path, and the response to thermal and geometric constraints.

• Utilize new theoretical understanding and predictive capability to (i) obtain a higher level of microstructural control, (ii) access new or more uniform growth structures, (iii) identify strategies to achieve new or better multiphase structures with novel or enhanced functionality. In addition to the control of alloy composition, growth velocity, and thermal gradient, avenues to ultimately explore will include novel thermal fields, geometric constraints, other applied fields, seeded growth, and techniques that require microgravity conditions.

SCOPE OF RESEARCH ACTIVITIES (includes both ground-based and microgravity experiments)

• Characterization and parameterization of the multi-scale 3D structure (phase distribution, interface structure, distribution of phase boundaries and orientation, eutectic fault distribution, domain structure, and larger scale grain structure);

• Quantification of natural directional growth transient (maze-lamellar transition, grain selection, fault formation and configuration, and fault domain geometry);

• Investigation of the response to forced thermal constraints (including off-axis thermal gradients that may be switched abruptly or rotated continuously, examining “locked” grain orientations and lock/unlock thresholds in 3D); and

• Investigation of response to forced geometric constraints (including off-axis growth from random and/or seed crystals, tilted and helical geometries, and geometric cross-sectional expansion or contraction).

COLLABORATION

While the proposed project is justifiable on its own merits, the work described here is also intended to provide collaborative support to the existing European Space Agency (ESA) project titled “Solidification along a Eutectic Path in Ternary Alloys” (SETA), which has made remarkable scientific progress in recent years. Facilitating open scientific interaction between the PI and the SETA team members (and between NASA and ESA), the project will be culminated with a set of flight-based microgravity experiments intended to validate the experimental correlations, mechanistic descriptions, and analytical models that are produced during the ground-based portion of this project and several other areas of investigation included within the SETA project. It is proposed that the planning and performance of flight-based solidification experiments will be done collaboratively, as will the analysis of post-solidification specimens upon their return to Earth.

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As mentioned in the Task Description, the objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled multiphase structures through advanced solidification processing. The potential Earth-based impact of this research is indeed far-reaching, given the extremely broad range of multiphase structures that can be fabricated through solidification processing, which is certainly not limited to metals and alloys. Indeed the properties of nearly any multiphase material can be improved by better control of multiphase structure. In addition, the fundamental and practical knowledge from this research may lead to innovative process technologies and new materials and/or structures.
Task Progress:	The technical progress made during this period (October 1, 2011 through September 30, 2012) includes research activities in the specific areas of (i) eutectic grain competition and selection behavior during directional solidification, (ii) directional growth structures and segregation patterns that form during univariant two-phase growth, and (iii) three-phase invariant solidification structures, all in the Al-Cu-Ag alloy system. Ground based research includes specific experiments aimed at the project science objectives listed above as well as other tests aimed at clarifying experimental techniques, growth conditions, and analysis methods to be used for the planned flight-based experiments. Technical progress in these areas is summarized briefly, below. INVARIANT BINARY TWO-PHASE GROWTH: We have investigated several aspects of coupled growth using binary Al-Cu as a test system, including grain selection, lamellar faults, grain boundary structure, topological ordering, and the various transient lengths that may be present in directional solidification. Year-2 results have elucidated many aspects of grain competition and correlations to ordering and lamellar spacing. We have also used binary alloys to examine the initial growth of a single-phase layer and other holding period and early stage structures that form during interrupted “stop and go” cyclic directional growth. Selected findings are listed below. (i) For directional growth velocities ranging from 0.0001 to 0.005 mm/s, grain selection occurs over a distance of 30-80 mm. (ii) The topological transient occurs within the first 5-10 mm of growth. During this period and for the first 15-20 mm of growth, the number of grains decreases rapidly, with well-ordered grains surviving. (iii) Well ordered grains compete over the remainder of the transient, with grain orientation and grain boundary structure being the most important factors. (iv) Interlamellar spacing tends to decrease as order increases. A change in lamellar spacing is observed as grains reach a peak cross sectional area and begin to shrink. (v) Expanding grains may exhibit decreased order behind the migrating boundary. (vi) Grain boundary migration mechanisms include the effects of faults, grain orientation, and topological order. UNIVARIANT TWO-PHASE GROWTH IN A TERNARY ALLOY: We have initiated the investigation of univariant coupled growth, employing specimens of Al-16.2at%Cu-3.6at%Ag and directional growth velocities of 0.0001, 0.0002, 0.0005, 0.0010, 0.0020, and 0.0050 mm/s with a nominal thermal gradient of 5-7 K/mm. Over this range, we have observed planar and cellular eutectic structures and have seen some evidence of dendritic structures. Detailed analyses of both lateral and axial chemical segregation is currently in progress, and these will be compared with computed solutal transient profiles for the univariant. Selected Year-2 findings are listed here. (i) A planar to nonplanar (cellular) transition in the two-phase growth front was observed to occur between 0.0005 and 0.001 mm/s. (ii) Lateral segregation pattern have been measured and found to be indicative of the cellular morphology. Work is now in progress to quantify both axial and lateral composition profiles (as a function of velocity) to facilitate analysis of interface stability and cellular evolution and comparison with theory. (iii) Evidence of significant convection has been observed in some cases, in the form of nonuniform structures. Work is now in progress to quantify the composition variation and to examine the effect of a reduced ampoule diameter. INVARIANT THREE-PHASE GROWTH IN A TERNARY ALLOY: Using alloys of ternary eutectic composition, directionally solidified at velocities from 0.0003 to 0.025 mm/s, we have investigated the three-phase structure of the ternary invariant growth mode. Simple eutectic spacings are not able to fully describe the structure, and suitable parameters for characterization of these structures have not been well-established. In Year 1, we developed an objective set of measurements to quantify the eutectic scale and morphology. We have examined these further in Year 2, and we have used the Fourier transform to assess ordering in these multiphase structures. We have investigated the influence of the quenching process on the observed structure, specifically looking at the near-quench structure, and will use serial milling in the coming year for this purpose. We will also begin to study orientation effects in detail and will continue our investigation of growth from seeds, where managing the single-phase layer growth and remelting will be important. Selected key findings are listed below. (i) Quantitative parameters such as phase spacing and intercept length provide measurable objective means to characterize complex three-phase structures. Such parameters will be essential to the development of comprehensive theories for three-phase growth. (ii) An accurate representation of the interface morphology can only be obtained ver near to the quenched interface. Additional serial milling work is currently being done to examinee the near-quench region. (iii) Grain orientation plays a major role in the selection of growth structure. It appears that dominant morphologies favor specific grain orientations, and we are currently quantifying these orientations. (iv) Three-phase structures exhibit complex fault structures.
Bibliography Type:	Description: (Last Updated: 06/01/2016)
Books/Book Chapters	Napolitano RE. "Univariant and invariant mulitphase solidification in ternary alloys." in "Solidification of containerless undercooled melts." Ed. D.M. Herlach, D.M. Matson. Hoboken, N.J. : Wiley-VCH, 2012. ISBN 978-3-527-33122-2., Aug-2012